



OCTOBER HIGH INSTITUTE
FOR ENGINEERING & TECHNOLOGY
ELECTRONICS AND COMMUNICATIONS
ENGINEERING DEPARTMENT

ENERGY-EFFICIENT LPWAN COMMUNICATION FRAMEWORK FOR SMART INFRASTRUCTURE MONITORING IN SMART CITIES



CONNECTING INFRASTRUCTURE.
ENABLING SMART DECISIONS.
BUILDING SUSTAINABLE CITIES.



ULTRA LOW
POWER



LONG RANGE
COVERAGE



SUPPORTS LARGE
NUMBER OF NODES



LOW COST
DEPLOYMENT



RELIABLE &
SECURE



INTRODUCTION

- The scale problem in smart cities requires a communication solution that is low-cost, long range, and energy efficient.
- This project proposes an energy-efficient LPWAN communication framework using LoRaWAN technology to monitor various smart infrastructures in real time and support data-driven decision making.



METHODOLOGY

- Sensors collect data from different infrastructures.
- Data is transmitted using LoRaWAN to a gateway.
- The gateway forwards data to the cloud server.
- The cloud processes and stores data for visualization.
- Users monitor and receive alerts through a dashboard.



KEY FEATURES



Long Range Communication



Ultra Low Power Consumption



Scalable Network



Cost Effective Solution



Real-Time Monitoring



High Reliability & Security



APPLIED APPLICATIONS

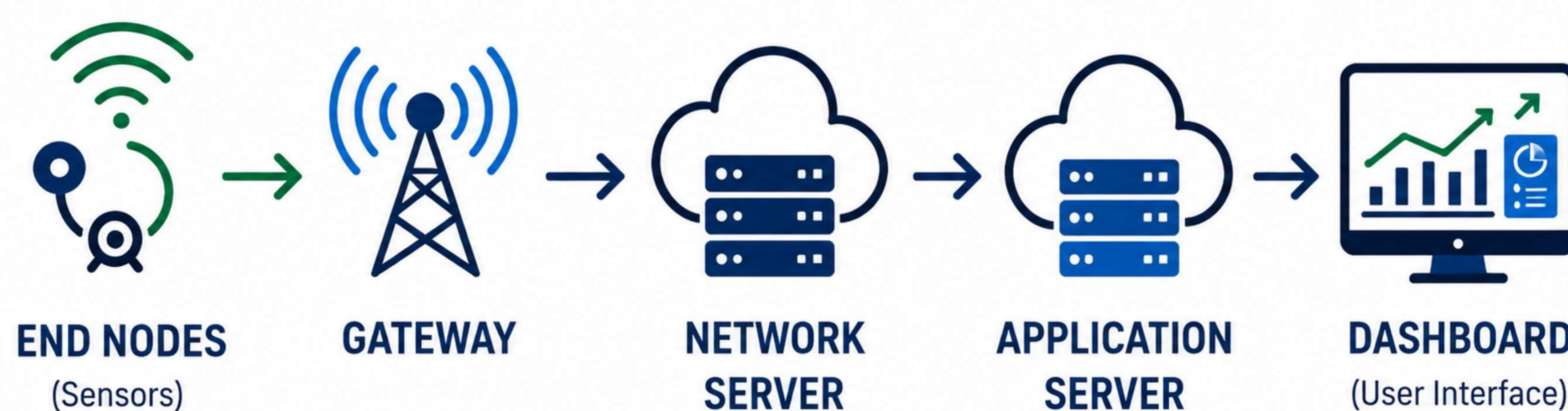
- Bridge Load Monitoring
- Soil Humidity Monitoring
- Soil Water Saturation Monitoring
- Gas Leakage Detection
- Water Leakage Detection
- Smart Lighting Monitoring



RESULTS

- Stable long range communication with high reliability.
- Low power consumption ensures long battery life.
- Real-time monitoring with instant alerts.
- Efficient data transmission with minimal latency.
- Scalable system for multiple applications.

SYSTEM ARCHITECTURE



HARDWARE IMPLEMENTATION



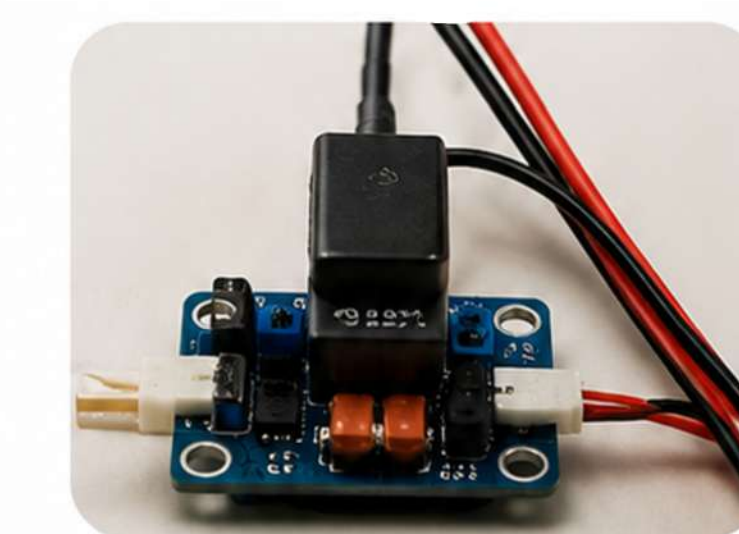
Load Cell
Weighing Monitoring



Gas Sensor
MQ-2



Soil Moisture Sensor
Capacitive



Water Level Sensor



Flow Sensor



Smart Lighting Node



CONCLUSION

The proposed LPWAN communication framework using LoRaWAN provides an efficient, scalable, and cost-effective solution for smart infrastructure monitoring in smart cities. It enables real-time data collection and analysis, helping in better resource management and decision making, leading to more sustainable and intelligent urban environments.



FUTURE WORK

- Deploying AI for predictive maintenance and anomaly detection.
- Developing a mobile dashboard for real time alerts and notifications.
- Integrating renewable energy sources for energy self-sufficiency.

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